Chapter 14, Problem 13P

Problem

Construct the Bode plots for

$$G(s) = \frac{0.1(s+1)}{s^2(s+10)}, \qquad s = j\omega$$

Step-by-step solution

Step 1 of 6

Consider the following transfer function.

$$G(s) = \frac{0.1(s+1)}{s^2(s+10)}$$

Substitute $j\omega$ for s in equation.

$$\begin{aligned}
G(\omega) &= \frac{0.1(1+j\omega)}{(j\omega)^2(10+j\omega)} \\
&= \frac{(0.1)(1+j\omega)}{(j\omega)^2(10)\left(1+\frac{j\omega}{10}\right)} \\
&= \frac{(0.01)(1+j\omega)}{(j\omega)^2\left(1+\frac{j\omega}{10}\right)} \\
&= \frac{\left(\frac{1}{100}\right)(1+j\omega)}{(j\omega)^2\left(1+\frac{j\omega}{10}\right)}
\end{aligned}$$

Step 2 of 6

Determine the Magnitude of the transfer function G_{dB}.

$$
\begin{aligned}
G_{dB} &= |G(\omega)| \\
&= 20\log_{10}\left|\frac{\left(\frac{1}{100}\right)(1+j\omega)}{(j\omega)^2\left(1+\frac{j\omega}{10}\right)}\right| \\
&= 20\left[\log_{10}\left(\left(\frac{1}{100}\right)|1+j\omega|\right) - \log_{10}|j\omega|^2\left|1+\frac{j\omega}{10}\right|\right] \\
&= 20\left[\{\log_{10}(0.01) + \log_{10}|1+j\omega|\} - \{2\log_{10}|j\omega| + \log_{10}\left|1+\frac{j\omega}{10}\right|\}\right]
\end{aligned}
$$

$$
\begin{aligned}
G_{dB} &= 20\log_{10}(0.01) + 20\log_{10}|1+j\omega| - 40\log_{10}|j\omega| - 20\log_{10}\left|1+\frac{j\omega}{10}\right| \\
&= -20\log_{10}(10)^2 + 20\log_{10}|1+j\omega| - 40\log_{10}|j\omega| - 20\log_{10}\left|1+\frac{j\omega}{10}\right| \\
&= -40 + 20\log_{10}|1+j\omega| - 40\log_{10}|j\omega| - 20\log_{10}\left|1+\frac{j\omega}{10}\right| \\
&= -40 - 40\log_{10}|j\omega| + 20\log_{10}|1+j\omega| - 20\log_{10}\left|1+\frac{j\omega}{10}\right|
\end{aligned}
$$

Hence, the magnitude of the transfer function G_{dB} is

$$\boxed{-40 - 40\log_{10}|j\omega| + 20\log_{10}|1+j\omega| - 20\log_{10}\left|1+\frac{j\omega}{10}\right| \text{ dB}}.$$

Step 3 of 6

Determine the Phase of transfer function ϕ.

$$
\begin{aligned}
\phi &= -180^\circ + \tan^{-1}(\omega) - \tan^{-1}\left(\frac{\omega}{10}\right) \\
&= -180^\circ + \tan^{-1}\left(\frac{\omega - \frac{\omega}{10}}{1+\omega\left(\frac{\omega}{10}\right)}\right) \\
&= -180^\circ + \tan^{-1}\left(\frac{\frac{9\omega}{10}}{\left(\frac{10+\omega^2}{10}\right)}\right) \\
&= -180^\circ + \tan^{-1}\left(\frac{9\omega}{10+\omega^2}\right)
\end{aligned}
$$

Hence, the Phase of transfer function ϕ is $\boxed{-180^\circ + \tan^{-1}\left(\frac{9\omega}{10+\omega^2}\right)}$.

Step 4 of 6

Determine the Phase angle ϕ, at various radiant frequency $\omega(\text{rad/s})$.

The following is the table 1, showing the phase angle ϕ, at various radiant frequency $\omega(\text{rad/s})$.

Table 1

$\omega(\text{rad/s})$	0.01	0.1	1	10	100	1000
ϕ	-179.48°	-174.86°	-140.71°	-140.71°	-174.86°	-179.48°

MATLAB code to plot Bode diagram:

```
>> num=[0.1 0.1];
>> den=[1 10 0 0];
>> x=tf(num,den);
>> bode(x)
>> grid
```


The following is the bode diagram.

The Bode diagram is shown in Figure 1.

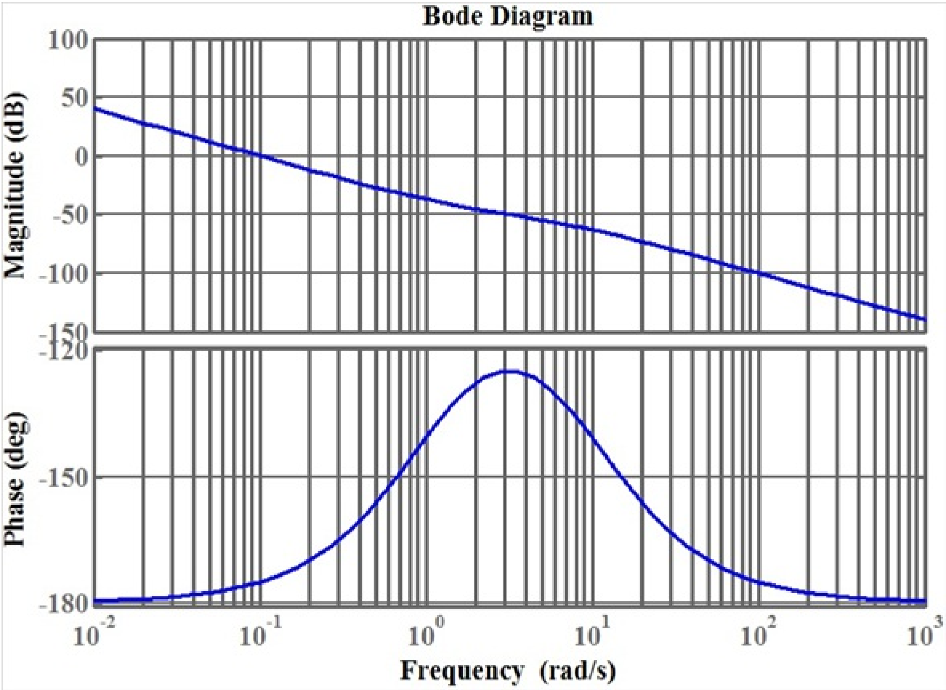

Figure 1

Hence, the bode plots for $G(s)$ has been $\boxed{\text{plotted}}$ as shown in Figure 1.